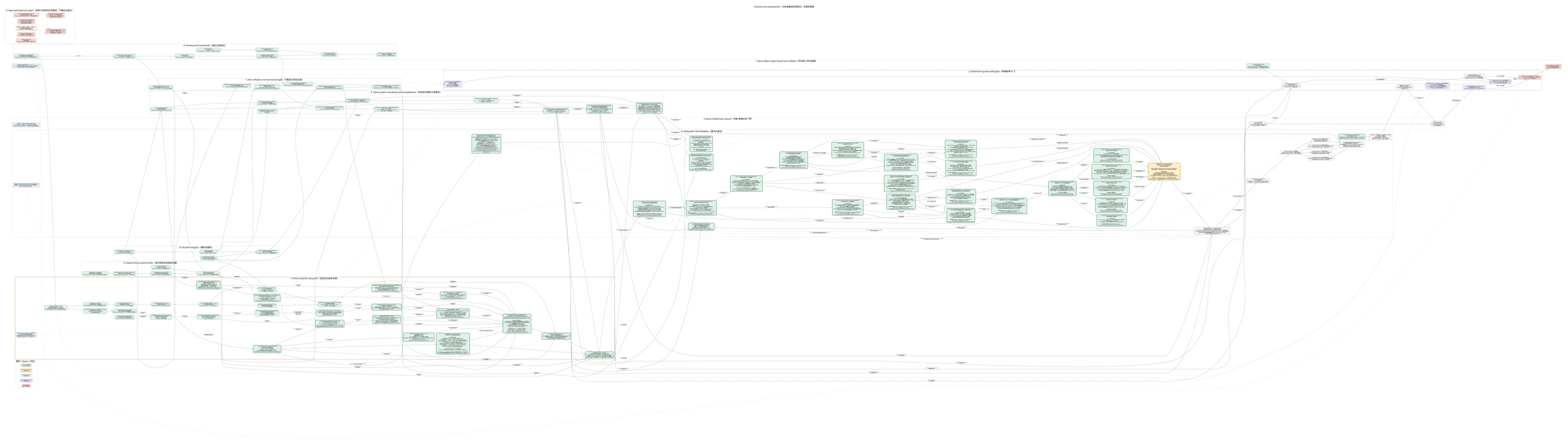

Checked min-proof growth（已检查最短证明增长）定理依赖图

K. Rejected historical routes（经审计弃用的历史路线；不通向主结论）
A. Honest proof coordinate（诚实证明坐标）
I. Same-object upper bound and collision（同对象上界与碰撞）
J. Submission-grade audit gates（投稿级审计门）
C. Non-inflation and canonical length（不膨胀与规范长度）
E. Same-system soundness and completeness（同系统可靠性与完备性）
H. Buss-Pudlák lower bound（布斯-普德拉克下界）
G. Arithmetic internalization（算术内部化）
B. Decoder integrity（解码完整性）
D. Explicit local costed verifier（显式带成本局部验证器）
F. Active total bit-cost proof（当前总位成本证明）
binaryNatCode / decodeBinaryNat
binaryTermCode + binaryFormulaCode
binarySequentCode
PAAxiomCertificate
StructuralValidityCertificate
CheckedPAProofTree.binaryCode
binaryCertifiedPAProofCode
packBinaryString / packedPayloadLength
Sondow checked certificates
Sondow same-object compiler
Podlák 1986 Theorem 3.1
Forbidden project assumptions（禁止项目级假设）
Final dependency audit（最终依赖审计）
Buss 1994 Theorem 5
PA consistency in the standard model
Consistency contradiction + exponent extraction
Bounded proof-calculus assembly
suitable numeration（合适枚举）
PA computation-trace theorem
CompactProofCx,t = Con_PA^compact(k)
Primitive-recursive numeric formula-list checks
Primitive-recursive formula-free token machine
Primitive-recursive formula-shift token machine
Primitive-recursive formula-negation token machine
Primitive-recursive direct formula constructors
Primitive-recursive term/formula/sequent value parsers
Primitive-recursive packed input layer
Primitive-recursive binary-Nat token stream machine
ListedCertifiedFiniteConsistencyAt(k)
Accepted-code bounded conjunction compiler
Accepted-code bounded modus ponens compiler
Primitive-recursive arithmetic term/formula parser
Exact arbitrary-input syntax inversion
Primitive-recursive closed-formula token machine
Primitive-recursive listed-proof structural parser
Primitive-recursive PA/structural certificate parsers
Exact noncanonical bit/token stream synchronization
Typed syntax decoder iff exact token prefix
Listed proof typed decoder iff exact token prefix
PA/structural certificate typed decoder iff exact token prefix
Primitive-recursive packed stream + exact typed-decoder equivalence
Exact arbitrary-input listed-proof inversion
Exact arbitrary-input PA/structural certificate inversion
Quantitative parameterized diagonalization
decodeBinaryNat_result
OptionTrace + traceBind
decodeManyVectorTrace_result
decodeCompactTermTrace_result
decodeCompactFormulaTrace_result
decodeCompactListedProofTrace_result
listedCompactCertifiedPAProofVerifierTrace_result
decodeBinaryNatTrace_success_failure_potential
decodeManyVectorTrace_potential
arithmeticFunc/Rel_decode_charge_le
decodeCompactTermTrace_parserCostBound
decodeCompactFormulaTrace_parserCostBound
decodeCompactListedProofTrace_parserCostBound
decodePAAxiomCertificateTrace_parserCostBound
10/10 listedCertificateValidTrace cost branches
packed proof/formula decoder cost bounds
Transformation binary-code length bounds
binaryFormulaNegTrace_result / _cost_le
Success/failure total potential bound
VerifierBitCost(x,y) <= p(|x|+|y|)
bitStringEqTrace / natEqTrace
bitStringMemTrace / SubsetTrace / SetEqTrace
formulaEq/Mem/Subset/SetEqTrace
listedCertificateValidTrace
decodeCompactTerm / Formula / Sequent
decodeCompactProof
decodePAAxiomCertificate
decodeCompactPackedCertifiedPAProof
shift/free/neg/substitution size bounds
decodeCompactListedProof
listedCompactCertifiedPAProofVerifier_eq_true_iff
listedCompactCertifiedPAProofVerifier_toDerivation
derivation_to_listedCompactCertifiedPAProofVerifier
compactCertifiedPAProofVerifier_canonical_length_bound
Concrete P(x,y) / ListedCompactCertifiedPAProofChecks
minListedCertifiedPAProofPayloadLength(formula)
P_canon(code, formulaCode) = canonical Bool verifier
Fforbid axioms audit（公理依赖审计）
图例（status / 状态）
Lean kernel checked Lean 内核已检查
Active proof obligation 当前证明义务
Pending root theorem 待完成根定理
Axiom/dependency audit 公理/依赖审计
Rejected route 经审计弃用路线